



THE COPPERBELT UNIVERSITY

MICHAEL CHILUFYA SATA SCHOOL OF MEDICINE

END OF TERM I TEST - JUNE, 2024

MBS 230 - MEDICAL BIOCHEMISTRY I

Student Identification Number (SIN) : DUBR

Program of Study: MBSCH

Duration of Examination : 02 hours

Total Marks : 80 marks

INSTRUCTIONS TO CANDIDATES

READ CAREFULLY BEFORE YOU START TO ANSWER

- Write your STUDENT IDENTIFICATION NUMBER (SIN) on the space provided on the front page.
- Read the instructions at the beginning of each section (Sections A and B). Answer all questions. No negative marking applies for wrong answers in Section A. Mark allocations for questions in Section B are shown in brackets.
- USE ONLY BLUE OR BLACK INK PEN to answer. Other colours may be used for diagrams only. Answers written in pencil shall not be considered, except if used for diagrams.
- Answer in this paper only. For section A, circle the letter corresponding to the correct answer and for section B, your answer should be in the provided space. Your answers should be LEGIBLE AND READABLE, otherwise they shall not be considered.
- No rough work is allowed anywhere in the paper, except on the last page (specifically allocated space). If any kind of rough work is found on any page, other than on the space specifically allocated, the page (s) shall not be considered.
- No one shall leave the examination room in the first 30 minutes and in the last 30 minutes of examination.

SECTION A: MULTIPLE CHOICE QUESTIONS - CHOOSE ONE THE CORRECT AND BEST ANSWER FOR EACH QUESTION AND CIRCLE THE LETTER IN FRONT OF IT. ANSWER ALL QUESTIONS (40 x 1 = 40 MARKS)

1. Dietary fatty acids are precursors for sphingolipids. Of the following, which one is formed from sphingolipids?
 - a. Lung surfactant
 - ☒ b. Myelin sheath ✓
 - c. Bile
 - d. Arachidonic acid
 - ☒ e. Blood lipoproteins
2. Low-dose aspirin is used as a prevention of platelet aggregation and heart attacks, whereas high-dose aspirin is used as an anti-inflammatory drug. Low-dose aspirin is used to block the formation of which eicosanoid?
 - a. Prostaglandins
 - b. Thromboxanes
 - c. Leukotrienes
 - ☒ d. Lysoxins
 - e. Epoxides
3. In humans, prostaglandins are derived primarily from which one of the following?
 - a. Glucose
 - b. Acetyl-CoA
 - ☒ c. Arachidonic acid ✓
 - ☒ d. Oleic acid
 - e. Leukotrienes
4. Some premature babies suffer from respiratory distress syndrome. Which class of lipids do these infants fail to secrete?
 - a. Triacyl glycerol
 - b. Sphingomyelin
 - ☒ c. Glycerophospholipids ✓
 - d. Glycolipids
 - e. Plasmalogens
5. Which of these enzymes is found in venoms of poisonous snakes and is responsible for the hydrolysis of fatty acids leading to formation of lysolecithin?
 - ☒ a. Phospholipase A2 ✓
 - b. Phosphatase A2
 - c. Lysase A2
 - d. Ligase A2
 - e. Isomerase A2

6. Which ether glycerolipid is produced in significant amounts in the fetus in late pregnancy and is responsible for stimulating the production of foetal lung surfactant?
- ☒ a. Platelet activating factor ✓
 - b. Plasmalogen
 - c. Sphingosine
 - d. Lipoprotein
 - e. Myelin
7. Which genetic disease arises when there is accumulation of sphingomyelin in the brain, liver and spleen and the condition is due to deficiency in sphingomyelinase enzyme?
- ☒ a. Nieman Pick ✓
 - b. Gaucher's
 - c. Tay Sachs
 - d. Glucocerebrosidosis
 - e. Sphingocerebrosidosis
8. Which of the following best describes the structure of a steroid?
- ☒ a. Three six-membered rings and one five membered ring all fused together ✓
 - b. Two six-membered rings and one five membered ring all fused together
 - c. Four five-membered rings and one five membered ring all fused together
 - d. One three-membered rings and one five membered ring all fused together
 - e. Three six-membered rings and one three membered ring all fused together
9. Which of the following steroids is a precursor of all animal steroids?
- a. Testosterone
 - b. Cortisol
 - c. Aldosterone
 - d. Estradiol
 - ☒ e. Cholesterol ✓
10. is an example of covalent modification used mainly for regulation of enzyme activity
- a. Amination
 - b. Acetylation
 - c. Phosphorylation
 - d. Methylation
 - e. Decarboxylation
11. Which of the following statements is correct about most enzymes?
- a. They increase the activation energy so that the reaction occurs
 - b. can make a reaction happen faster
 - c. are used once and need to be replaced after catalyzing a reaction
 - d. they are stable in both high acid and high alkaline solutions
 - e. are not specific to the type of substrate they receive
12. Which one of the following statements concerning glutamine is correct?
- a. Contains an amide group
 - b. Has Glu as its three letter abbreviation
 - c. Classified as an acidic amino acid
 - d. Has a charged polar side chain
 - e. Doesn't have a chiral carbon

13. Succinate dehydrogenase catalyses the dehydrogenation of succinate to fumarate. Malonic acid is used to inhibit the action of this enzyme. What type of inhibition is this?
- Competitive
 - Uncompetitive
 - Non competitive
 - Feed back
 - Allosteric
14. is associated with a condition known as phenylketonuria.
- Proline
 - Methionine
 - Phenylalanine
 - Histidine
 - Valine
15. An amino acid that yields acetoacetyl CoA during catabolism of its carbon skeleton is.....
- Glucogenic
 - Ketogenic
 - Hydrophobic
 - Polar
 - Polar and hydrophobic
16. is an amino acid that contains an imidazole ring
- Tyrosine
 - Phenylalanine
 - Histidine
 - Proline
 - Tryptophan
17. A 50-year-old patient is brought to the emergency department with the diagnosis of myocardial infarction. Which lactate dehydrogenase (LDH) isoenzyme activity would prevail in the patient's serum in the first two days after hospitalisation?
- LDH₁
 - LDH₂
 - LDH₃
 - LDH₄
 - LDH₅
18. A chiral carbon of an amino acid arises from the fact that it.....
- is symmetric
 - is bonded to both amino and carboxylic groups
 - is bonded to four different chemical groups
 - has net charge of zero
 - assumes an L configuration

19. The stability of protein structures is supported by a number of bonds. Which bonds support the primary structure?
- Ionic
 - Peptide
 - Hydrogen
 - Hydrophilic
 - Hydrophobic
20. The normal hemoglobin of an adult person consists of two identical α and two identical β polypeptide chains. What is this level of structure called?
- Primary
 - Secondary
 - Quaternary
 - Tertiary
 - Binary
21. One of the common laboratory experiments in Biochemistry determines electrophoretic spectrum of plasma proteins. What is the physiochemical properties of proteins utilized in this method?
- Inability to denaturation
 - Presence of charge
 - Optical activity
 - Solubility
 - Viscosity
22. Amino acids are an important substrate for gluconeogenesis during fasting. Which one of the following is a major gluconeogenic amino-acid received by the liver?
- Alanine
 - Glycine
 - Cysteine
 - Pyruvate
 - Tryptophan
23. Which amino acid is involved in producing glutathione, an important antioxidant?
- Glutamine
 - Glutamate
 - Proline
 - Ornithine
 - Arginine
24. A protein's activity is altered when a particular serine side chain is phosphorylated. Which of the following amino acid substitutions at this position could lead to a permanent alteration in normal enzyme activity?
- Serine to Glutamic acid
 - Serine to Threonine
 - Serine to Tyrosine
 - Serine to Lysine
 - Serine to Leucine

25. One of the main sources of non-volatile acid in the body is sulfuric acid generated from the sulfur-containing compounds in ingested food or from the metabolism of the sulfur-containing amino acids. Which of the following amino acids would lead to sulfuric acid formation?
- Cysteine and isoleucine
 - Cysteine and alanine
 - Cysteine and methionine
 - Methionine and isoleucine
 - Isoleucine and alanine
26. What would be the most likely composition of amino acids that form a portion of a transmembrane protein located within the plasma membrane?
- acidic
 - basic
 - hydrophobic
 - glycosylated
 - acetylated
27. Five isoenzymic forms of lactate dehydrogenase from the human blood serum were isolated and their properties studied. What property indicates that the isoenzymic forms were isolated from the same enzyme?
- Same physiochemical properties
 - Catalyzation of the same reaction
 - The same electrophoretic mobility
 - Tissue localisation
 - Same molecular weight
28. Which one of the following can influence the charge of amino acid residues in the active site of an enzyme?
- Competitive inhibitor
 - Excess substrate
 - pH of medium
 - Temperature
 - Atmospheric pressure
29. Regulation of enzyme activity is the basis for feedback inhibition in enzyme catalysis. Which one of the following represents the most common type of regulation?
- Covalent modification
 - Allosteric regulation
 - Proteolysis
 - Carboxylation
 - Hydroxylation

30. Which of these is an example of the secondary structure of a protein
- Peptide bonding between amino acids
 - Hydrophobic interactions
 - Hydrogen bonding between R groups
 - Disulphide bonds between cysteine residues
 - Hydrogen bonding an amine and carbonyl group
31. Which level of protein structure is determined only by hydrogen bonds
- Primary
 - Tertiary
 - Secondary
 - Quaternary
 - Ovoidal
32. Which globin chains are affected in glycosylated Hemoglobin?
- γ -globin chains
 - α -globin chains
 - β -globin chains
 - δ -globin chains
 - both γ and α -globin chains
33. Patients, homozygous for hemoglobin C, have a condition that arises from.....
- substitution of lysine by glutamic acid in the β -chain
 - substitution of glutamic acid by lysine in the β -chain
 - substitution of valine by glutamic acid in the β -chain
 - substitution of a polar amino acid by a non-polar one in the β -chain
 - substitution of one of the α -chains by a β -chain
34. Which one of these statements is correct about Thalassémias?
- hereditary hemolytic diseases in which there is an imbalance in the synthesis of globin chains
 - they are polygenic disorders
 - only α -globin chains are affected
 - they are not in a class of hemoglobinopathies
 - Only γ -chains are affected
35. Which two amino acids in collagen usually undergo post translational modification?
- lysine and histidine
 - glycine and alanine
 - valine and isoleucine
 - lysine and proline
 - leucine and glutamic acid
36. Lack of post translational modifications of some amino acids in collagen leads to non-cross linkage of the fibers. Which one of the following refers to the condition that arises?
- sickle cell anemia
 - pernicious anemia
 - scurvy
 - phenylketonuria
 - hemosiderosis

37. Which of the following is responsible for specifying the 3-dimensional shape of a protein?
- a. The peptide bond
 - b. The amino acid sequence
 - c. Interaction with other polypeptides
 - d. Interaction with molecular chaperons
 - e. Interaction with water molecules
38. Which one of these would not be considered as a function of lipids
- a. Energy storage
 - b. Structural components of cell membranes
 - c. Protection of internal organs
 - ☒ d. Help in absorption of water soluble vitamins. ✓
 - e. Act as detergent agents
39. Which one of these will not be considered as derived lipids
- a. Terpenes
 - b. Steroids
 - ☒ c. Carotenoids ✓
 - d. Male sex hormones
 - e. Tocopherols
40. Which one of these lipid compounds have choline part structure?
- a. Sphingosine
 - ☒ b. Lecithin ✓
 - c. Gerniol
 - d. Triacylglycerol
 - e. Phosphatidic acid

SECTION B: DESCRIPTIVE ANSWER TYPE/SCENARIO BASED QUESTIONS.
ANSWER ALL QUESTIONS. FOR EACH ANSWER, ILLUSTRATIONS WITH
DIAGRAMS (WHERE POSSIBLE) WILL ATTRACT MAXIMUM MARKS (40
MARKS)

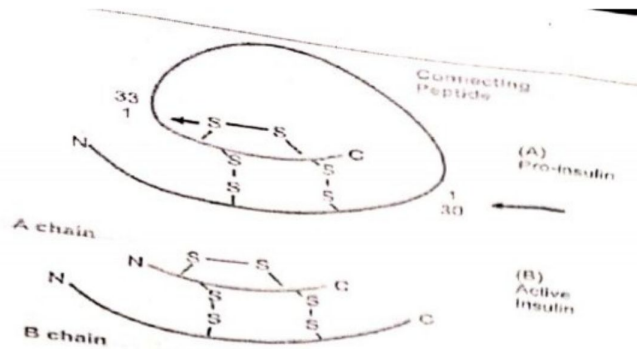
1. Plasma proteins are composed mainly of glycoproteins, lipoproteins and immunoglobulins that function in the immune system. These plasma proteins are involved in binding and transporting nutrients, toxic substances etc.
 - i. What is the normal reference range for total plasma proteins? (1 mark)
.....
 - ii. What is the term given to plasma that lacks coagulation factors? (1 mark)
.....
 - iii. How are normal reference ranges for biochemical parameters established? (2 marks)
.....
.....
.....
 - iv. Which one is a major plasma protein and what is its percentage concentration (1 mark)
.....
 - v. Name at least one function of this protein (1 mark)
.....
 - vi. Which plasma protein is elevated during inflammation and infections? (1 mark)
.....
 - vii. What is its classification? (1 mark)
.....
 - viii. Where are most plasma proteins synthesized? (1 mark)
.....
2. A new drug is developed which selectively cleaves covalent bonds between two sulfur atoms of non-adjacent amino acids in a polypeptide chain.
 - i. Which level of protein structure in affected molecules would be most directly affected by the drug? (1 mark)

ii. Explain why this level is affected (1 mark)

iii. Describe the other levels of protein structure (4 marks)

iv. Which forces or interactions maintain levels of protein structures. Note, for each level indicate the force/interaction (2 marks)

3. Beta cells of the pancreas synthesize insulin as a prohormone. Proinsulin is a single polypeptide chain with 86 amino acids. Biologically active insulin (2 chains) is formed by removal of the central portion of the pro-insulin before release.



For questions below refer to the figure above

- i. Why does the pancreas produce insulin as a prohormone? (1 mark)
.....
- ii. How is the prohormone converted to an active hormone? (1 mark)
.....
- iii. What happens to the connecting polypeptide indicated in the figure? (1 mark)
.....
- iv. What is the general term for enzymes that cut at the sites indicated by arrows? (1 mark)
.....
- v. Which amino acid is responsible for producing the S-S linkages in the figure? (1 mark)
.....
- vi. Explain the fact that species variation in insulin is restricted to position 8, 9, 10 in the A chain and to position 30 in the B chain (3 marks)
.....
.....
.....
- vii. What is the full name of a peptide formed by carboxyl group of glycine with amino group of Alanine and then carboxyl group of Alanine with amino group of Valine? (1 mark)
.....
.....

4. Polyunsaturated fatty acids (PUFAs) are essential fatty acids required by the body and cannot be synthesized and so they have to be included in diet



Refer to the figure above

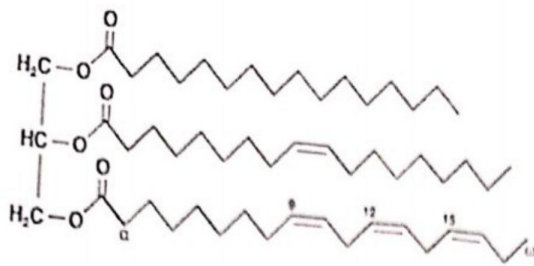
The figure above shows a saturated fatty acid and an unsaturated fatty acid

- Which one of the two has a higher melting point? (1 mark)
Saturated fatty acid.
- What would be the effect on the melting point if there is an increase in the number of double bonds in the unsaturated fatty acid above? (1 mark)
melting point decreases with the number of double bonds hence it will decrease.
- Name one powerful locally acting metabolite whose precursor is PUFAs (1 mark)
eicosanoids

Linoleic acid can be denoted as 18:2; ω -6 ^{$\Delta^9,12$}

- Define this notation (1 mark)
18 carbons, 2 double bonds, position 6
- Arachidic acid can be denoted as 20:0. Define this notation (1 mark)
✓
- Arachidonic acid is semi essential fatty acid. Explain why it is classified as such (1 mark)
✓

5. Refer to the figure below as you answer the questions below



The figure above is a presentation of a Triacylglycerol

- i. What are the two products of hydrolysis of the above Triacylglycerol? (1 mark)
glycerol and fatty acid
- ii. Name two tissues where these products are metabolised (1 mark)
muscles and liver
- iii. Give two reasons as to why triacylglycerols are highly concentrated stores of energy (2 marks)
exists in reduced form
anhydrous form
- iv. How much energy is realised from complete oxidation of a gram of a fatty acid? (1 mark)
9 kcal/g (38 kJ/g)
- v. How many carbon atoms are realised per cycle in: (2 marks)
 Beta oxidation... 2-C as acetyl CoA
 Alpha oxidation... 1-C in the form of CO₂
- vi. When does omega oxidation become active? (1 mark)
in conditions of impaired Beta oxidation
- vii. How many fatty acid chains are in the structure above? (1 mark)
3

END